

Executive Summary:

What English-Tuned Abliteration Misses in Slovene

September 2026

1 Goal

This study asks whether abliteration—removing refusal behaviour by orthogonalising weights against an English-extracted “refusal direction”—transfers to Slovene. Two sibling 12B models from the Gemma-3 family were compared: `google/gemma-3-12b-it` (English-centric) and `cjvt/GaMS3-12B-Instruct` (Slovene continual-pretraining variant), both under NF4 quantisation. The study ran five iterations with pre-registered predictions, independent audits, and 17 falsified hypotheses.

2 What Was Tried

- **Matched Heretic abliteration** (116 TPE trials per model, identical seed) to produce abliterated checkpoints, scored bilingually with keyword markers and LLM judges.
- **Refusal-direction transfer screen**: 2×2 source-language $\times$ evaluation-language ablation matrix, increment tests, and matched-efficacy controls.
- **Surrogacy panels** (exp6, exp7): 200+ Heretic edits per model, measuring whether English refusal outcomes predict Slovene outcomes.
- **Causal mechanism tests** (exp8): activation steering and weight-edit arms to isolate the Slovene residual.
- **Coverage $\times$ strength factorial** (exp9, exp10): 122 and 57 cells testing whether depth coverage or total edit energy governs Slovene refusal survival.
- **Causal write-profile measurement** (exp13, exp14): single-layer edits at all 48 layers, plus an energy-normalised overlap instrument O .
- **Gradient-blind fraction** (exp15): quantifying Heretic’s keyword objective’s structural blindness.
- **Corrected objective** (exp11): replacing keywords with a partial-aware classifier and re-running selection.
- **Cross-model replication** (exp12): Qwen3-8B and Mistral-7B across EN/SL/DE/LT.
- **Three independent audit/recompute passes** (eval1–eval4) checking 400+ numbers.

3 Key Results

1. **GaMS3’s edit works bilingually; Gemma’s does not.** GaMS3’s ablation reduced refusal from 100% to ~6% in both EN and SL. Gemma’s edit reduced EN refusal from 99% to 70% but left SL refusal at 95% (T1, S4 held-out, judged).
2. **The gap is judge-sensitive.** On S5X paired translations (Gemma edit), the EN–SL gap ranges from +0.06 (keyword) to +0.69 (strict judge), with the broad definition at +0.23. The PARTIAL class (54.8% of Gemma edit EN outputs) is the wedge.
3. **Residual refusal depends on edit placement, not only on total energy.** At matched total removal energy AND matched layer count, where in depth the edit is applied predicts how much refusal survives—in both languages, in both sibling checkpoints, and in Qwen3-8B. This means evaluating refusal-edit transfer across languages requires controlling for edit placement, not only total edit strength. Two one-forward-pass baselines (O_{cos} , O_{band4}) match or beat the expensive causal instrument at predicting residual refusal.
4. **The activation-space depth index fails cross-model.** Spearman = -0.009 over 21 model-language-weight cells. Two baselines beat both the index and the geometric predictor.
5. **The keyword objective is cross-lingually biased.** $\Delta_{\text{lang}} = -1.56$ $[-1.68, -1.44]$: the English keyword rule over-reports in English (FP rate 0.78) and is silent in Slovene (FN rate 0.82). Threshold blindness fraction = 1.0 in both searches.
6. **Utility is unaffected.** All macro utility changes were ≤ 0.20 points; FLORES+ NLL shifts were ≤ 0.002 nats/token.

4 What Failed

Seventeen hypotheses were falsified (F1–F17). The most consequential:

- **Depth coverage explains the gap** (F1): $\Delta R^2 = 0.040$, LOO -0.002 .
- **Broad-and-weak beats narrow-and-strong** (F2): falsified opposite (-0.10).
- **Corrected objective produces a better edit** (F5): dose-scaling the keyword edit achieved comparable gap reduction at lower KL.
- **r_prior direction carries Slovene residual** (F7): causal cut $0.036 < \text{best random control } 0.074$.
- **Count-based depth index predicts cross-model** (F3): Spearman -0.009 .
- **Anchor search is blinder overall** (F17): paired GBF difference $+0.006$; judge-referenced reverses.

5 Best Result

The firm positive is a **matched-budget placement contrast**: at fixed total edit energy and fixed layer count, the depth placement of the edit predicts how much refusal survives. Matched-energy edits at different placements produced Slovene strict refusal rates ranging from 0.27 to 0.92

depending on placement alone. This held for Gemma, GaMS3, and Qwen3-8B. At the refusal floor of shipped edits, however, dose—not placement—is the active variable ($p = 0.774$). The implication for cross-lingual refusal evaluation is that placement is a confound: a cross-lingual gap measured at one placement may not generalise to another, so evaluation protocols that compare refusal across languages must control for edit placement, not only total edit strength.

On the guard side, the official RefusEU ASR (Llama-Guard-3-8B + PolyGuard agreement) for the Gemma edit is EN 0.738 vs SL 0.103 = +0.635 gap, confirming the behavioural asymmetry with an independent safety metric.

6 Open Questions

1. **Native-speaker validation.** All Slovene material is machine-translated (Gemini-2.5-flash); five packets (640 items) await native-speaker labelling.
2. **Slovene judge certification.** The SL judge gate was not met ($\kappa = 0.723 < 0.80$); all SL gap claims are JUDGE_SENSITIVE.
3. **NF4 confound.** Paired bf16 vs NF4 gap: bf16 +0.60 vs NF4 +0.45 ($n = 20$, not resolved). The asymmetry is at least as large in bf16.
4. **Two checkpoints $\neq$ a population.** Nothing is attributed to continual pretraining, instruction tuning, or any training stage.
5. **Optimizer variance.** One seed per model; prompt-level CIs do not capture run-to-run optimizer variance.
6. **Placement at the floor.** The placement contrast is strongest mid-range; at shipped-edit refusal levels, dose dominates ($p = 0.774$ for placement swap).
7. **Defect rate.** 136 numbers independently re-derived; defect rate 15.4% [10.3%, 22.5%] on previously unaudited sections. No sign reversals.